

Title: What Should We Play Next? A Board Game Recommender for Groups Using BoardGameGeek Ratings

Dataset

BoardGameGeek reviews (Kaggle: [jvanelteren/boardgamegeek-reviews](https://www.kaggle.com/jvanelteren/boardgamegeek-reviews)), on Jolie at `/mnt/data/public/bgg/`. Ratings from `bgg-19m-reviews.csv` (user, ID, rating; comments unused): 18,964,794 ratings after dropping 13 below 1, from 412,815 users across 21,839 games, with no duplicate user-game pairs. After keeping users with at least 20 ratings: 169,027 users and 17,613,867 ratings. Player counts and other attributes come from `games_detailed_info.csv` (21,631 games). 213 rated games have no metadata, and 173 games (0.8%) have an unknown minimum or maximum player count and are excluded from player-filtered results.

Problem

A group of friends often can't decide which board game to try next, and browsing 20,000+ titles doesn't help. Given the games a group has played, which ones they liked and disliked, and how many players they are, can we recommend games they haven't played that they're likely to enjoy? A group is treated as one user who pools their ratings.

Solution abstract

Keep users with at least 20 ratings and split them into training and held-out users. Train ALS matrix factorization on the training users and compare it against a popularity baseline and item-item CF. For each held-out user, give the model part of their ratings (explicit low ratings count as dislikes, unrated games as unknown) and rank unrated games by folding the user into the learned game factors. Filter results to games whose stated minimum and maximum player counts include the group's size. Evaluate with Precision@10, Recall@10, and NDCG@10 (relevant = held-out rating of 8 or above, fixed in advance), plot accuracy against the number of games the user enters, and report how much the player-count filter (tested at 2, 3, 4, and 5 players) reduces NDCG. Optional: frequent itemsets of co-liked games to explain recommendations.